

ASSIGNMENT #1

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
Professor: Dr. Michael Caracotsios
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Your name: ERICA CARUSO

Due Date: Wednesday January 22 by 5:00pm in my office

Number of points.....100

Your score.....61

Note: All problems are worth 100 points. Your final score will be the average value

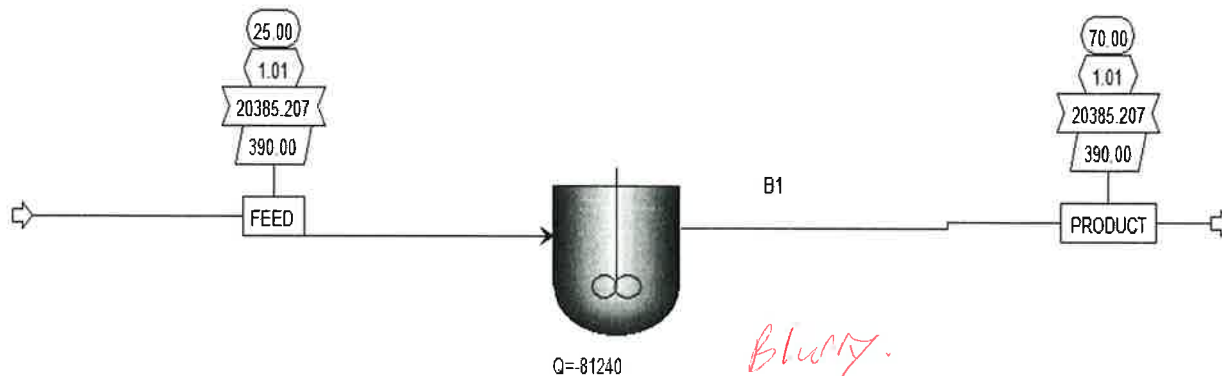


Figure 1. ASPEN PFD

Blurry.

Units - D

Heat and Material Balance Table			
Stream ID		FEED	PRODUCT
Temperature	C	25.0	70.0
Pressure	bar	1.01	1.01
Vapor Frac		0.000	0.000
Mole Flow	kmol/hr	390.000	390.000
Mass Flow	kg/hr	20385.207	20385.207
Volume Flow	l/min	365.211	390.194
Enthalpy	MMBtu/hr	-135.514	-136.674
Mole Flow	kmol/hr		
ACETI-01		193.000	70.245
ETHAN-01		187.000	64.245
WATER		10.000	132.755
ETHYL-01			122.755

Why does this look weird?

Very poorly formatted.

Figure 2. Heat and Weight Balance

Not a figure. This is a table and the label goes above the table.

- 1

Calculations:

	FEED	PRODUCT
Density (kg/m ³)	930.2943 ✓	870.7301 ✓
Heat Capacity (J/g*K)	1.9125 ✓	2.5465 ✓
Heat of Reaction (J/mol)	-366,356.10	-369,493.68
Equilibrium Constant	14.58967	
Water Flow Rate (kg/hr)	17547.78 ✓	
Heat Transfer Coeff. (J/min*K)	452777.76 ✓	

Table 1. Calculations from ASPEN

Density, Heat Capacity, Heat of Reaction and Water flow rate were all given in the results in ASPEN. Water flow rate was found by using the Utility option in the simulation. ASPEN asked for a heating/cooling value for the water and 30 BTU/lb was entered in. This outputted the water flow rate of the cooling water as a utility. There was a calculation performed to find the heat transfer coefficient (UA). The formula to find UA is shown below:

$$Q_{cool} = UA(T_{in} - T_R)$$

The Q term is the duty calculated by ASPEN and for this simulation it was calculated as 81239.74 cal/sec. This can be converted to Joule/second which would equal 339582.109. Temperature of the feed is 50C and the temperature of the reactor is 70C. Plugging in Q_{cool} , T_{in} , and T_R into the equation, UA can be calculated. The equilibrium constant also had to be calculated from the information ASPEN gave from the simulation. The equation to solve for the equilibrium constant is shown below. The gamma term was found using the ASPEN results.

$$K_{eq} = \Pi(x_i \gamma_i)^{v_i}$$

x_i = mole fraction

γ_i = liquid activity coefficient

v_i = stoichiometric coefficient

	x_i	γ_i	v_i	K
Acetic Acid	0.182	0.622	-1	8.83361
Ethanol	0.165	1.341	-1	4.519468
Water	0.341	2.068	1	0.705188
Ethyl Acetate	0.315	1.687	1	0.531405

$$K_{eq} = 14.58967$$

Table label?

Discussion?

Derived Equations:

Acetic Acid Balance:

$$V \frac{dC_{AA}}{dt} = QC_{AAfeed} - QC_{AA} + r_{AA}V$$
$$r_{AA} = -R_1 = -k_f C_{AA} C_{ETOH} + k_r C_{EAC} C_{WATER}$$
$$U(1) := C_{AA}$$

Ethanol Balance:

$$V \frac{dC_{ETOH}}{dt} = QC_{ETOHfeed} - QC_{ETOH} + r_{ETOH}V$$
$$r_{ETOH} = -R_1 = -k_f C_{AA} C_{ETOH} + k_r C_{EAC} C_{WATER}$$
$$U(2) := C_{ETOH}$$

Ethyl Acetate Balance:

$$V \frac{dC_{EAC}}{dt} = QC_{EACfeed} - QC_{EAC} + r_{EAC}V$$
$$r_{EAC} = +R_1 = k_f C_{AA} C_{ETOH} - k_r C_{EAC} C_{WATER}$$
$$U(3) := C_{EAC}$$

Water Balance:

$$V \frac{dC_{WATER}}{dt} = QC_{WATERfeed} - QC_{WATER} + r_{WATER}V$$
$$r_{WATER} = +R_1 = k_f C_{AA} C_{ETOH} - k_r C_{EAC} C_{WATER}$$
$$U(4) := C_{WATER}$$

Energy Balance:

$$\rho C_p V \frac{dT}{dt} = Q \rho C_p (T_{feed} - T) + (-\Delta H_{R1}) R_1 V - UA(T - T_w)$$
$$U(5) = T$$

work needed.

-1

Title

Parameters	Athena Variable	Description and Units
$Q = 365.21$	Q	Volumetric Flow Rate, L/min
$T_{\text{feed}} = 298$	T_{feed}	Feed Temperature, K
$C_{\text{AAfeed}} = 8.81, C_{\text{ETOHfeed}} = 8.81, C_{\text{EACfeed}} = 0.0, C_{\text{WATERfeed}} = 8.81$	$C_{\text{AAfeed}}, C_{\text{ETOHfeed}}, C_{\text{EACfeed}}, C_{\text{WATERfeed}}$	Feed Composition, mol/L
$\text{Vol} = 500$	Vol	Reactor Volume, L
$UA = 814997.1$	UA	Heat Transfer Coefficient, J/(min*K)
$T_w = 323$	T_w	Cooling Water Temperature, K
$\rho = 870.73$	ρ	Reaction Mixture Density, g/L
$C_p = 1.9125$	C_p	Heat Capacity, J/(g*K)
$\Delta H_{\text{R1}} = -10000$	ΔH_{R}	Heat of Reaction, J/mol
$k_{f0} = 1.9 \times 10^8$	k_{f0}	Pre-exponential factor, Kmol/(m ³ *s)
$k_{r0} = 5.0 \times 10^7$	k_{r0}	Pre-exponential factor, Kmol/(m ³ *s)
$E_{f0} = 59500$	E_{f0}	Activation Energy, J/mol
$E_{r0} = 59500$	E_{r0}	Activation Energy, J/mol

Add another column for values.
- 1

CODE:

!Variable Declarations and Data

!*****

Parameter Rg=8.3145 As Real !Universal Gas Constant, J/(mol K)

Global Q,Tfeed,CAAFed,CETOHfeed,CEACfeed,CWATERfeed As Real

Q=365.21 !L/min

Tfeed=298 !K

CAAFed=8.81 !mol/L

CETOHfeed=8.81 !mol/L

CEACfeed=0.0 !mol/L

CWATERfeed=8.81 !mol/L

Global Vol,UA,Tw As Real

Vol=500 !L

UA=814997.1 !J/min K

Tw=323 !K

Global rho,Cp,DHR1 As Real

rho=870.7301 !g/L

Cp=1.9125 !J/g K

DHR1=-10000 !J/mol

Global kfo,kro,Er As Real

kfo=1.9E8 !kmol/(m³*s)

kro=5.0E7 !kmol/(m³*s)

Er=59500 !KJ/mol

@Initial Conditions

U(1)=4.405

U(2)=0.0

U(3)=0.0

$$U(4)=4.405$$

$$U(5)=343$$

@Model Equations

Dim CAA,CETOH,CEAC,CWATER, Temp As Real

$$CAA=U(1)$$

$$CETOH=U(2)$$

$$CEAC=U(3)$$

$$CWATER=U(4)$$

$$Temp=U(5)$$

Dim kf,kr,R1 As Real

$$kf=kfo*\exp(-Er/(Temp*Rg))$$

$$kr=kro*\exp(-Er/(Temp*Rg))$$

$$R1=kf*CAA*CETOH-kr*CEAC*CWATER$$

$$\text{If}(iRange==1)Tw=318.0$$

$$\text{If}(iRange==2)Tw=323.0$$

$$F(1)=Q*CAA_{feed}-Q*CAA+(-R1)*Vol$$

$$F(2)=Q*CETOH_{feed}-Q*CETOH+(-R1)*Vol$$

$$F(3)=Q*CEAC_{feed}-Q*CEAC+(R1)*Vol$$

$$F(4)=Q*CWATER_{feed}-Q*CWATER+(R1)*Vol$$

$$F(5)=Q*\rho*Cp*(T_{feed}-Temp)+(-DHR1)*R1*Vol-UA*(Temp-Tw)$$

@Coefficient Matrix

$$E(1)=Vol$$

$$E(2)=Vol$$

$$E(3)=Vol$$

$$E(4)=Vol$$

$$E(5)=\rho*Vol*Cp$$

@Solver Options

Neq=5

Npts=50

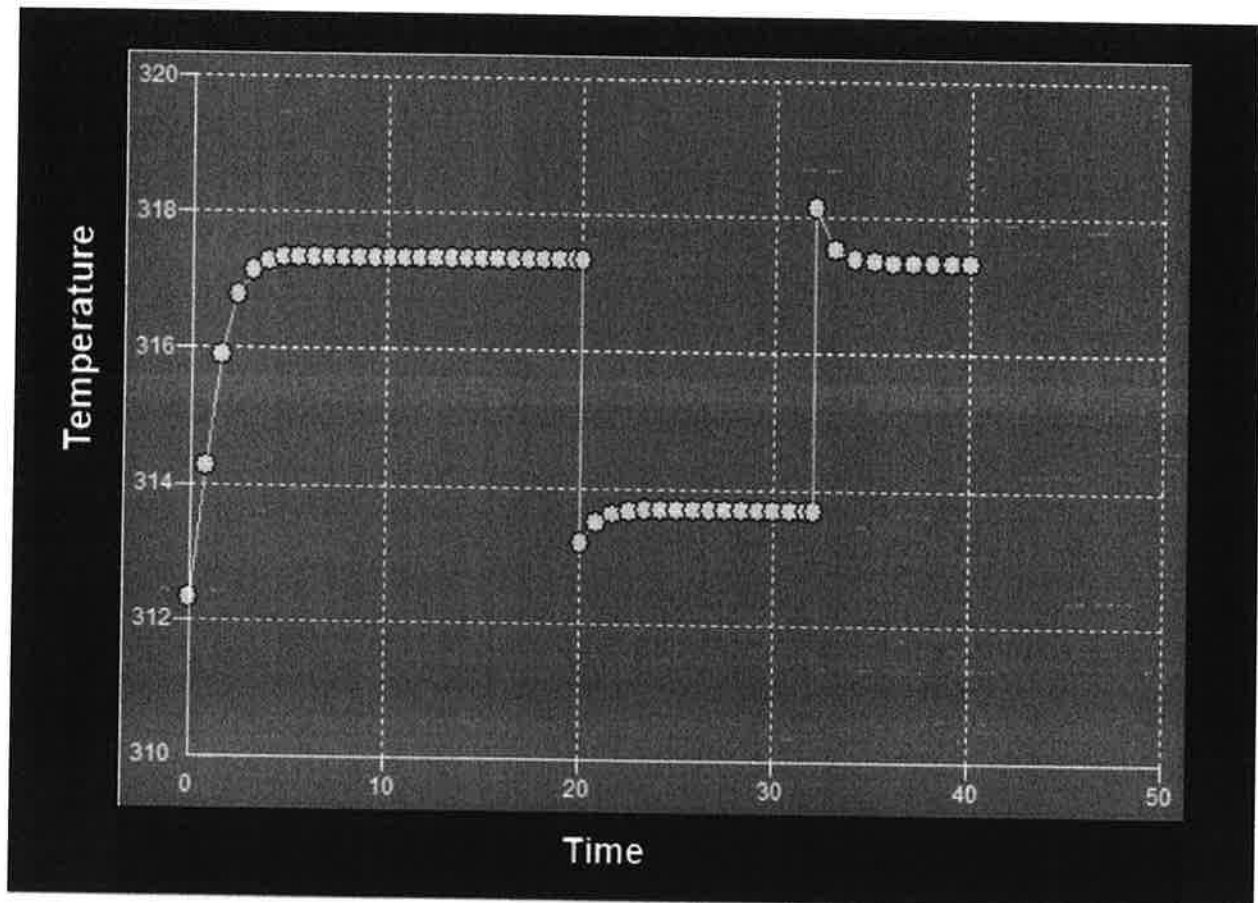
Tend=40

Tstop=20;32

Parameter(28)=1

VariableNames=Time;CAA;CETOH;CEAC;CWATER;Temperature

GRAPH:



7.
Na - 15 units, 2

